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**Java SE 27 & JDK 27**

# Class FilterOutputStream

java.lang.Object
    java.io.OutputStream
        java.io.FilterOutputStream

**All Implemented Interfaces:**

Closeable, Flushable, AutoCloseable

**Direct Known Subclasses:**

BufferedOutputStream, CheckedOutputStream, CipherOutputStream, DataOutputStream, DeflaterOutputStream, DigestOutputStream, InflaterOutputStream, PrintStream

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```
public class FilterOutputStream
extends OutputStream
```

This class is the superclass of all classes that filter output streams. These streams sit on top of an already existing output stream (the *underlying* output stream) which it uses as its basic sink of data, but possibly transforming the data along the way or providing additional functionality.

The class `FilterOutputStream` itself simply overrides all methods of `OutputStream` with versions that pass all requests to the underlying output stream. Subclasses of `FilterOutputStream` may further override some of these methods as well as provide additional methods and fields.

**Since:**

1.0

## *Field Summary*

**Fields**

| Modifier and Type | Field | Description |
|---|---|---|
| protected OutputStream | **out** | The underlying output stream to be filtered. |

## *Constructor Summary*

**Constructors**

| Constructor | Description |
|---|---|
| **FilterOutputStream**(OutputStream out) | Creates an output stream filter built on top of the specified underlying output stream. |

## *Method Summary*

| All Methods | Instance Methods | |
|---|---|---|

| Modifier and Type | Method | Description |
|---|---|---|
| `void` | `close()` | Closes this output stream and releases any system resources associated with the stream. |
| `void` | `flush()` | Flushes this output stream and forces any buffered output bytes to be written out to the stream. |
| `void` | `write(byte[] b)` | Writes `b.length` bytes to this output stream. |
| `void` | `write(byte[] b, int off, int len)` | Writes `len` bytes from the specified `byte` array starting at offset `off` to this output stream. |
| `void` | `write(int b)` | Writes the specified `byte` to this output stream. |

**Methods declared in class OutputStream**

`nullOutputStream`

**Methods declared in class Object**

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

## *Field Details*

### out

```
protected OutputStream out
```

The underlying output stream to be filtered.

## *Constructor Details*

### FilterOutputStream

```
public FilterOutputStream(OutputStream out)
```

Creates an output stream filter built on top of the specified underlying output stream.

**Parameters:**

`out` - the underlying output stream to be assigned to the field `this.out` for later use, or `null` if this instance is to be created without an underlying stream.

## *Method Details*

### write

```
public void write(int b)
           throws IOException
```

Writes the specified `byte` to this output stream.

The `write` method of `FilterOutputStream` calls the `write` method of its underlying output stream, that is, it performs `out.write(b)`.

Implements the abstract `write` method of `OutputStream`.

**Specified by:**

`write` in class `OutputStream`

**Parameters:**

`b` - the `byte`.

**Throws:**

IOException - if an I/O error occurs.

## write

```
public void write(byte[] b)
           throws IOException
```

Writes b.length bytes to this output stream.

**Overrides:**

write in class OutputStream

**Implementation Requirements:**

The write method of FilterOutputStream calls its write method of three arguments with the arguments b, 0, and b.length.

**Implementation Note:**

Note that this method does *not* call the one-argument write method of its underlying output stream with the single argument b.

**Parameters:**

b - the data to be written.

**Throws:**

IOException - if an I/O error occurs.

**See Also:**

write(byte[], int, int)

## write

```
public void write(byte[] b,
                  int off,
                  int len)
           throws IOException
```

Writes len bytes from the specified byte array starting at offset off to this output stream.

**Overrides:**

write in class OutputStream

**Implementation Requirements:**

The write method of FilterOutputStream calls the write method of one argument on each byte to output.

**Implementation Note:**

Note that this method does not call the write method of its underlying output stream with the same arguments. Subclasses of FilterOutputStream should provide a more efficient implementation of this method.

**Parameters:**

`b` - the data.

`off` - the start offset in the data.

`len` - the number of bytes to write.

**Throws:**

`IOException` - if an I/O error occurs.

`IndexOutOfBoundsException` - If `off` is negative, `len` is negative, or `len` is greater than `b.length - off`

**See Also:**

`write(int)`

## flush

```
public void flush()
           throws IOException
```

Flushes this output stream and forces any buffered output bytes to be written out to the stream.

**Specified by:**

`flush` in interface `Flushable`

**Overrides:**

`flush` in class `OutputStream`

**Implementation Requirements:**

The `flush` method of `FilterOutputStream` calls the `flush` method of its underlying output stream.

**Throws:**

`IOException` - if an I/O error occurs.

**See Also:**

`out`

## close

```
public void close()
           throws IOException
```

Closes this output stream and releases any system resources associated with the stream.

**Specified by:**

`close` in interface `AutoCloseable`

**Specified by:**

`close` in interface `Closeable`

**Overrides:**

`close` in class `OutputStream`

**Implementation Requirements:**

When not already closed, the `close` method of `FilterOutputStream` calls its `flush` method, and then calls the `close` method of its underlying output stream.

**Throws:**

`IOException` - if an I/O error occurs.

**See Also:**

`flush()`, `out`

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Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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